

PROJECT DEVELOPMENT PHASE

User Creation

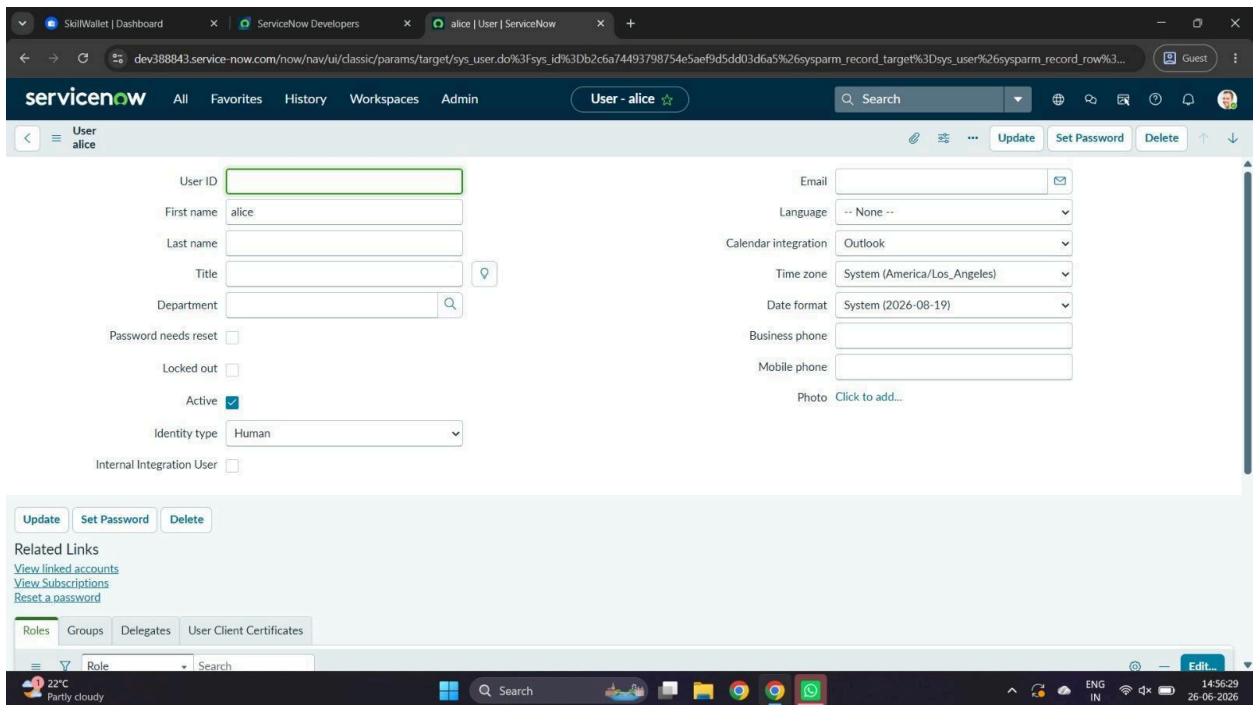
Objective

The objective of this phase is to create user accounts in ServiceNow that will participate in the workflow and access control process. These users represent different members of the organization and are assigned appropriate roles based on their responsibilities.

Explanation

In this project, two users named Alice and Bob were created in the ServiceNow instance. These users are used to test role assignments, group membership, workflow execution, and approval processes. Creating separate users helps simulate a real organizational environment where different employees perform different tasks.

User Creation – Alice



The screenshot shows the ServiceNow user creation interface for a user named 'alice'. The form is divided into two main sections: user details and settings. The user details section includes fields for User ID, First name (alice), Last name, Title, Department, Password needs reset, Locked out, Active (checked), Identity type (Human), and Internal Integration User. The settings section includes fields for Email, Language, Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (2026-08-19)), Business phone, Mobile phone, and Photo. Below the form are buttons for 'Update', 'Set Password', and 'Delete'. There are also links for 'View linked accounts', 'View Subscriptions', and 'Reset a password'. At the bottom, there are tabs for 'Roles', 'Groups', 'Delegates', and 'User Client Certificates'. The bottom of the screen shows a Windows taskbar with the date and time 14:56:29 on 26-06-2026.

User ID:

First name:

Last name:

Title:

Department:

Password needs reset: ☐

Locked out: ☐

Active: ☒

Identity type:

Internal Integration User: ☐

Email:

Language:

Calendar integration:

Time zone:

Date format:

Business phone:

Mobile phone:

Photo: [Click to add...](#)

[Update](#) [Set Password](#) [Delete](#)

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

[Roles](#) [Groups](#) [Delegates](#) [User Client Certificates](#)

Role:

Search:

22°C Partly cloudy

14:56:29 26-06-2026

The Alice user account was created successfully and is used for testing user management and access permissions within the application.

User Creation – Bob

The screenshot shows the ServiceNow user profile page for a user named 'bob'. The page is divided into two main sections: personal information on the left and system settings on the right. The left section includes fields for User ID (Bob), First name (bob), Last name, Title, Department, Password needs reset, Locked out, Active (checked), Identity type (Human), and Internal Integration User. The right section includes fields for Email, Language (None), Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (2026-08-19)), Business phone, Mobile phone, and Photo (Click to add...). Below the form are buttons for 'Update', 'Set Password', and 'Delete'. At the bottom, there are links for 'Related Links' (View linked accounts, View Subscriptions, Reset a password) and tabs for 'Roles (2)', 'Groups (2)', 'Delegates', and 'User Client Certificates'. The bottom of the page shows a Windows taskbar with various icons and a system tray with weather and time information.

The Bob user account was created successfully and is used as the assigned user in workflow testing and approval scenarios.

Group Creation

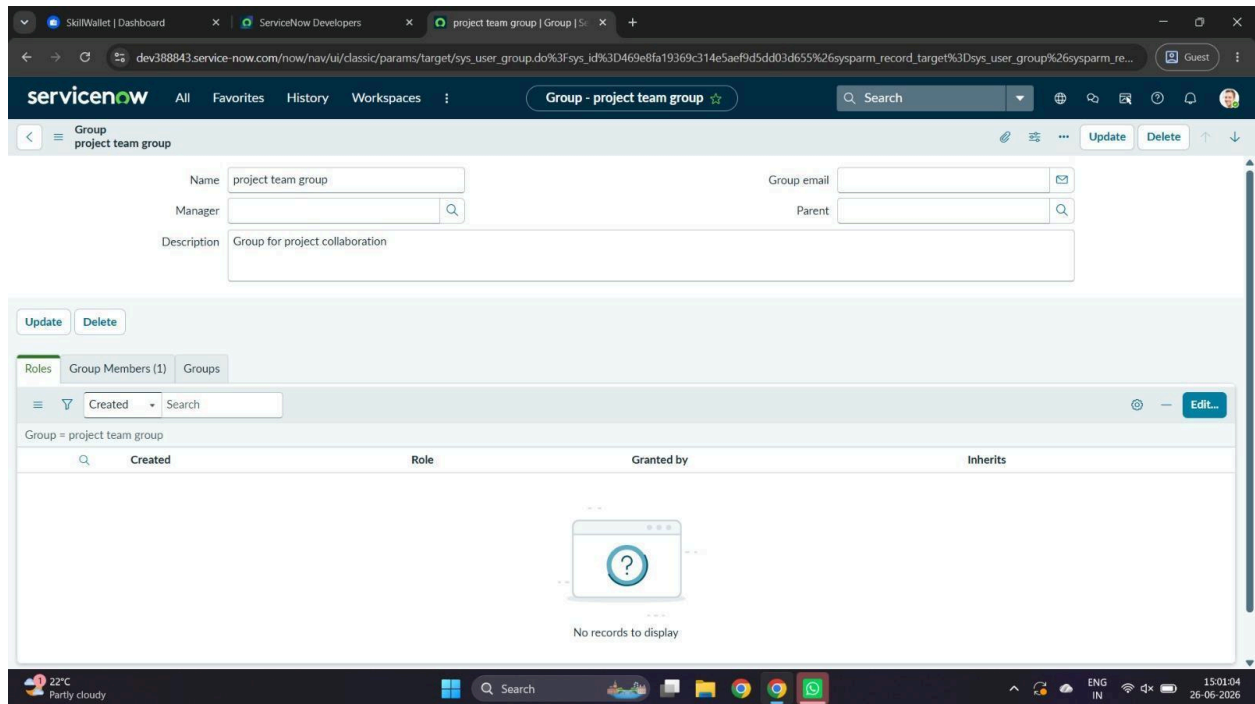
Objective

To organize users into a common group for simplified management and role assignment.

Explanation

A group named Project Team Group was created in ServiceNow. This group represents users working on the same project. By assigning users to a group, permissions can be managed more efficiently, reducing the need to assign roles individually.

Project Team Group Creation



The Project Team Group was created successfully, and users can now be assigned to this group for centralized access management.

Role Creation

Objective

To define roles that determine user responsibilities and system permissions.

Explanation

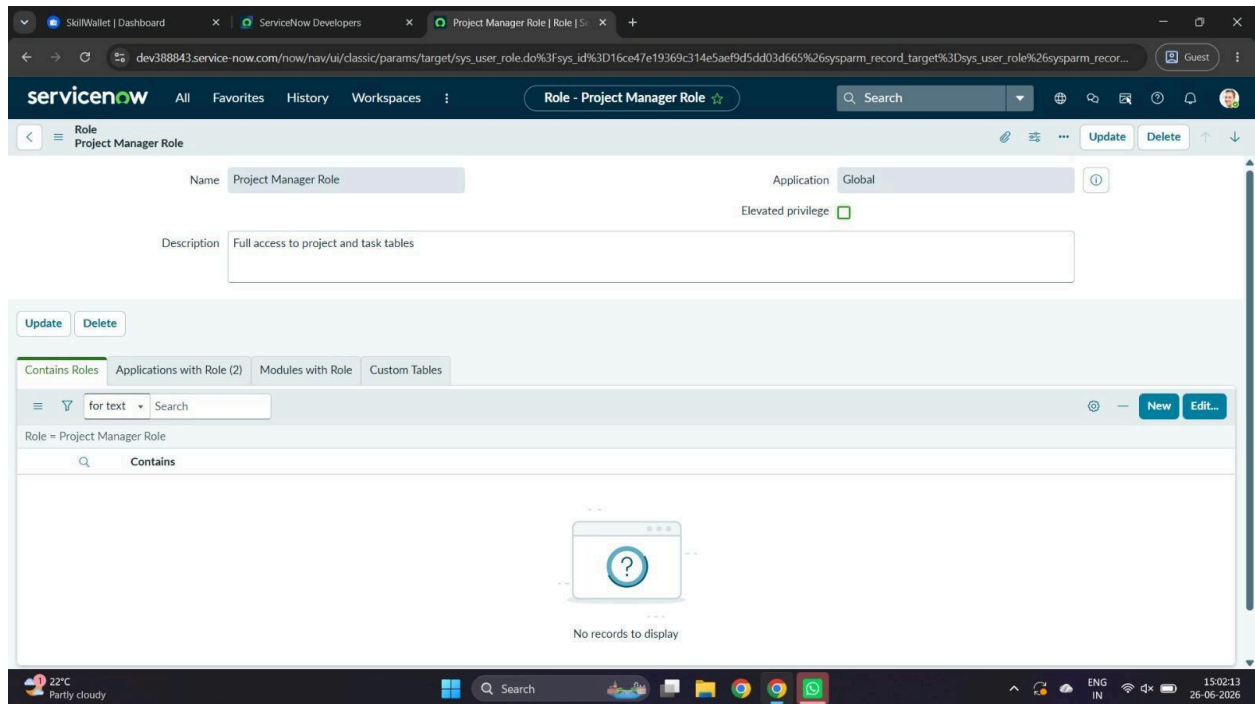
Two custom roles were created:

Project Manager Role

Team Member Role

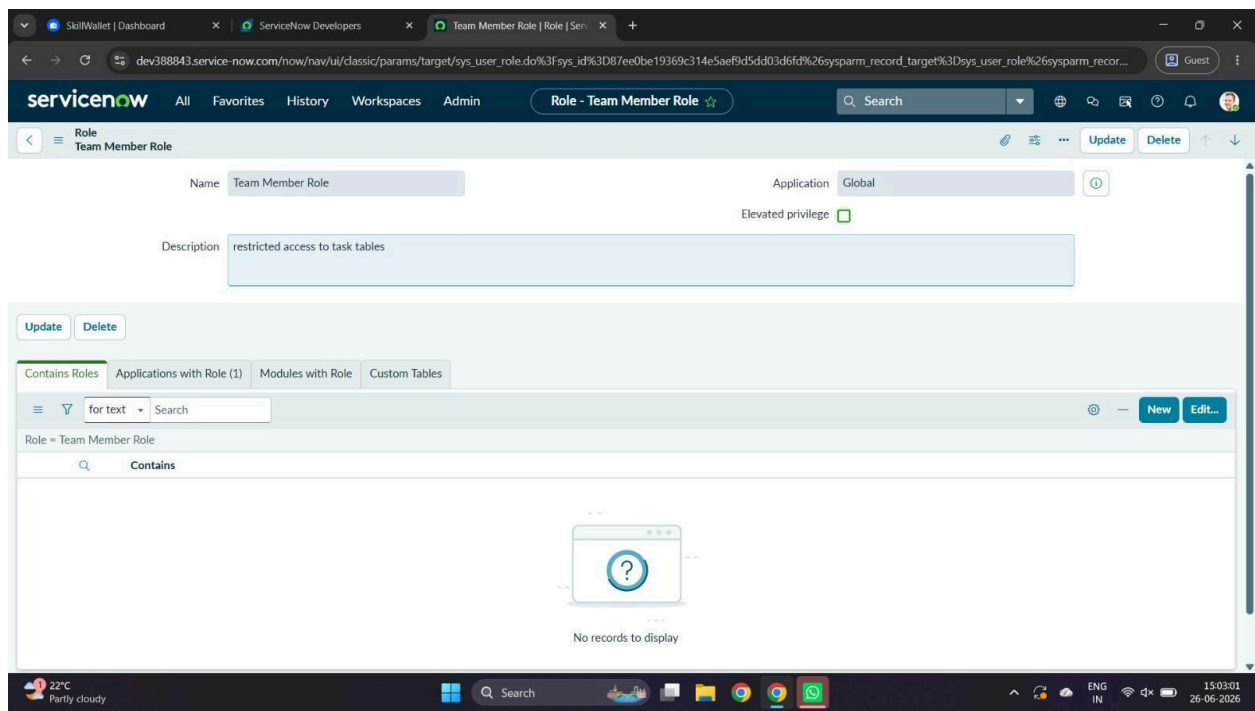
These roles help implement Role-Based Access Control (RBAC) by ensuring users receive only the permissions required to perform their tasks.

Project Manager Role



The Project Manager Role provides higher-level permissions required to manage project-related activities.

Team Member Role



The Team Member Role provides limited permissions required for team members to perform assigned tasks securely.

Assigning Roles to Users

Objective

To assign appropriate roles to users based on their responsibilities.

Explanation

After creating users and roles, the roles were assigned to the respective users. This ensures that users can access only the features and records permitted by their assigned role.

Role Assignment

Role assignment enables secure access control by providing users with permissions based on their organizational responsibilities.

Objective

To create custom tables for storing project and task information.

Explanation

Two custom tables were developed:

Project Table

Task Table 2

These tables store project details and task information required for workflow automation and approval management.

Project Table Creation

dev388843.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26sysparm_checked_items%3D%26syspar...

serviceNow All Favorites History Workspaces Admin Table - New Record Search

Table New record Submit Cancel

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label project table

* Name u_project_table

Extends table

Application Global

Create module ☒

Create mobile module ☒

Add module to menu -- Create new --

New menu name project table

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
X	project name	String				false
X	project manager	String				false
X	due date	Date				false
X	status	String				false
+	Insert a new row...					

22°C Mostly cloudy 15:09:12 26-06-2026

Task Table 2 Creation

dev388843.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D80719b299369c314e5ae9d5dd03d625%26sysparm_record_target%3Dsys_db_object%26sysparm_rec...

serviceNow All Favorites History Workspaces Admin Table - Task Table 2 Search Delete Update Delete All Records

Table Task Table 2

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Task Table 2

* Name u_task_table_2

Application Global

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search 1 to 11 of 11 New

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
	Created	Date/Time	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
X	comments	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
X	description	String (Full UTF-8)	(empty)	50		false
X	task name	String (Full UTF-8)	(empty)	20		false
X	status	String	(empty)	40		false

T-storms In the morning 15:10:33 26-06-2026

Assigning Roles to Project Table and Task Table

Objective

The objective of this phase is to associate the appropriate roles with the Project Table and Task Table 2 so that only authorized users can access and perform operations on these tables.

Explanation

After creating the custom tables, role-based permissions were configured. The Project Manager Role and Team Member Role were assigned to the respective tables to ensure secure access. This implementation follows the Role-Based Access Control (RBAC) model, where users can access only the data and functions permitted by their assigned roles.

Assigning roles to tables improves security by preventing unauthorized users from creating, modifying, or viewing sensitive project information.

Project Table with Assigned Role

Assigning Project Manager Role to Project Table

The screenshot shows the ServiceNow 'Application Menu - Project Table' configuration page. The page is titled 'Application Menu - Project Table' and includes a search bar and navigation links. The main configuration area contains the following fields and options:

- Title:** Project Table
- Application:** Global
- Active:** ☒
- Roles:** Project Manager Role
- Category:** Custom Applications
- Hint:** (empty text box)
- Description:** (empty text box)

At the bottom of the configuration area, there are 'Update' and 'Delete' buttons. Below the configuration area, there is a 'Modules' section with a search bar and a 'New' button. The bottom of the screen shows a Windows taskbar with the date and time 15:20:55 on 26-06-2026.

The Project Table was configured with the Project Manager Role to ensure that only authorized users can manage project records.

Task Table 2 with Assigned Role

Assigning Team Member Role to Task Table 2

The screenshot shows the ServiceNow 'Application Menu - Task Table 2' configuration page. The page includes a title field set to 'Task Table 2', an 'Application' dropdown set to 'Global', and an 'Active' checkbox that is checked. A 'Roles' section lists 'u_task_table_2_user, Project Manager Role, Team Member Role'. The 'Category' is set to 'Custom Applications'. There are also fields for 'Hint' and 'Description'. At the bottom, there are 'Update' and 'Delete' buttons. The browser's address bar shows the URL: dev38843.service-now.com/now/nav/ui/classic/params/target/sys_app_application.do%3Fsys_id%3Dc5d197699369c314e5aef9d5dd03d6a8%26sysparm_record_target%3Dsys_app_application%26s...

The Task Table 2 was configured with the Team Member Role, allowing authorized team members to access and update task records securely.

Access Control (ACL) Configuration

Objective

The objective of this phase is to implement Access Control Lists (ACLs) to protect sensitive information and restrict access based on user roles.

Explanation

Access Control Lists (ACLs) were created to provide field-level security for the Task Table 2. ACLs ensure that only users with the appropriate role can create, view, or modify specific fields. This prevents unauthorized access and strengthens the overall security of the application. The following ACLs were configured:

Status

Comments

Assigned To

Description

Status ACL

The screenshot shows the 'Access Control - New Record' form in ServiceNow. The form is for creating a new record. The 'Type' is set to 'record', 'Operation' is 'create', and 'Application' is 'Global'. The 'Active' checkbox is checked, and 'Admin overrides' is also checked. The 'Protection policy' is set to 'None --'. The 'Name' is 'Task Table 2 [u_task_table_2]' and the 'status' field is selected. The 'Description' field is empty. A warning message at the top states: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' Below the form, the 'Conditions' section is expanded, showing a list of roles: 'Team Member Role'. The 'Requires role' section is also expanded, showing the same role. The bottom of the screen shows the Windows taskbar with the date and time as 15:29:16 on 26-06-2026.

The Status field ACL ensures that only authorized users can create or modify the task status.

Comments ACL

The screenshot shows the 'Access Control - New Record' form in ServiceNow, similar to the previous one but for a 'Comments ACL'. The 'Type' is 'record', 'Operation' is 'create', and 'Application' is 'Global'. The 'Active' checkbox is checked, and 'Admin overrides' is also checked. The 'Protection policy' is set to 'None --'. The 'Name' is 'Task Table 2 [u_task_table_2]' and the 'comments' field is selected. The 'Description' field is empty. The 'Conditions' section is expanded, showing a list of roles: 'Team Member Role'. The 'Requires role' section is also expanded, showing the same role. The bottom of the screen shows the Windows taskbar with the date and time as 15:31:53 on 26-06-2026.

The Comments field ACL restricts access, ensuring that only users with the required role can update task comments.

Assigned To ACL

Access Control - New Record

* Type: record

* Operation: create

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: Task Table 2 [u_task_table_2]

assigned to: assigned to

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Team Member Role

Insert a new row...

The Assigned To ACL allows only authorized users to assign tasks to team members.

Description ACL

The screenshot shows the 'Access Control - New Record' form in the ServiceNow interface. The form is for creating a new record. Key fields include:

- Type:** record
- Operation:** create
- Admin overrides:** ☒
- Protection policy:** -- None --
- Name:** Task Table 2 [u_task_table_2]
- Description:** (empty text area)
- Application:** Global
- Active:** ☒
- Advanced:** ☐

Below the form, there is a 'Conditions' section with a blue box containing text about Access Control Rules. Below that is a 'Requires role' section with a table showing the role 'Team Member Role'.

Role
Team Member Role

The bottom of the screenshot shows the Windows taskbar with the date and time 15:34:55 on 26-06-2020.

The Description ACL protects project information by restricting unauthorized modifications.

Workflow Development using Flow Designer

Objective

The objective of this phase is to automate the project approval process using ServiceNow Flow Designer.

Explanation

A workflow was created using Flow Designer to automate task processing. The workflow begins automatically when the specified trigger conditions are met. It updates the task record, requests approval, and changes the task status based on the approval outcome.

Automation eliminates repetitive manual work, improves efficiency, and ensures that approvals follow a standardized process.

Workflow Trigger

The screenshot shows the ServiceNow Workflow Studio interface for a workflow named 'task table'. The 'TRIGGER' section is active, showing a configuration for 'Task Table 2 Created where (status is in progress, and comments is feedback, and assigned to is bob)'. The trigger is set to 'Created' and the table is 'Task Table 2 [u_task_table_2]'. The conditions are: 'status' is 'in progress', 'comments' is 'feedback', and 'assigned to' is 'bob'. The 'Data' panel on the right shows the flow variables for the trigger, including 'Task Table 2 Record', 'Task Table 2 Table', 'Run Start Time UTC', and 'Run Start Date/Time'. The 'ACTIONS' section is currently empty.

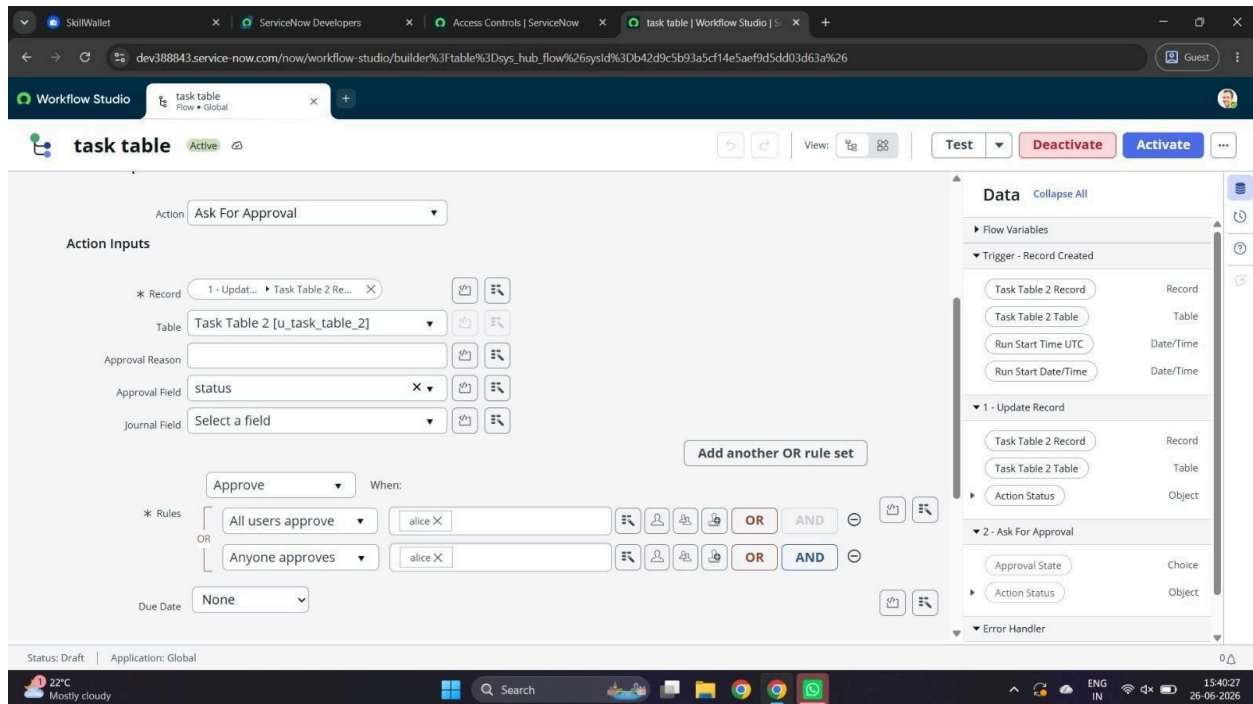
The workflow is triggered automatically when the defined conditions are satisfied, initiating the approval process.

Update Record Action

The screenshot shows the ServiceNow Workflow Studio interface for the same workflow. The 'ACTIONS' section is active, showing a configuration for 'Update Task Table 2 Record'. The action is set to 'Update Record'. The 'Action Inputs' section shows the record source as 'Trigger - R... Task Table 2 R...', the table as 'Task Table 2 [u_task_table_2]', and the fields to be updated as 'status' with the value 'Completed'. The 'Data' panel on the right shows the flow variables for the action, including 'Task Table 2 Record', 'Task Table 2 Table', 'Run Start Time UTC', and 'Run Start Date/Time'. The 'ACTIONS' section is currently empty.

The Update Record action modifies the task information automatically before the approval request is initiated.

Ask for Approval Action



The Ask for Approval action sends the task to the designated approver for review and approval.

Testing and Validation

Objective

The objective of this phase is to verify that the developed application functions correctly and meets the project requirements.

Explanation

Testing was conducted by creating sample task records and executing the workflow. Each component of the application, including user management, role assignment, ACLs, and workflow automation, was verified to ensure proper functionality.

The workflow executed successfully, approvals were processed correctly, and task records were updated automatically based on the approval results.

Update Record Testing

ServiceNow interface showing the 'Task Table 2s' list. The table displays task details including assigned to, comments, description, status, and task name. The status column shows 'Completed', 'requested', and 'approved'.

assigned to	comments	description	status	task name
Search	Search	Search	Search	Search
bob	feedback		Completed	
bob	feedback		requested	
bob	feedback		approved	

1 to 3 of 3

The Update Record action executed successfully, confirming that task details were updated automatically.

Approval Workflow Testing

ServiceNow interface showing the 'Approvals' list. The table displays approval details including State, Approver, Comments, Approval for, and Created. The State column shows 'Approved'.

State	Approver	Comments	Approval for	Created
Approved	alice		(empty)	2026-06-24 21:43:58

Number of rows removed from this list by Security constraints: 19

1 to 20 of 745

The approval workflow was executed successfully, and the task status was updated according to the approval decision.